AI REVENUE MANAGER

for Holiday Homes

Execution PRD + Agent Architecture | Version 2.0

Lyzr Manager–Worker Framework | Initial Market: Dubai, UAE

Field	Value
Document Type	Execution PRD + Agent Architecture
Version	2.0
Build Framework	Lyzr Manager–Worker (Agentic)
Initial Market	Dubai, UAE
Scope	V1 Build with V2/V3 Forward References

PART 1: STRATEGIC CONTEXT

1.1 The Problem Worth Solving

Dubai's short-term rental market is one of the highest-yielding and highest-volatility property markets globally. DTCM-licensed properties operating on Airbnb, Booking.com, Vrbo, and Homes & Villas face a unique set of conditions that make manual revenue management both time-consuming and costly:

- Demand is fragmented across 20+ feeder markets (GCC, India, UK, Russia, CIS, China) each with independent seasonality curves and booking lead times
- Event-driven demand spikes are extreme and brief — GITEX, New Year's Eve on The Palm, Dubai World Cup, and Art Dubai compress availability across entire sub-markets within days
- Competitor reaction speed is asymmetric — large operators with automated tools move prices in hours; manual operators lose the window entirely
- LOS patterns are non-standard — corporate relocations drive 14–30 night bookings; leisure guests book 3–5 nights; optimal strategy changes by unit type, zone, and lead time
- The PMS ecosystem is fragmented — Hostaway, Guesty, Lodgify, Hostfully, Beds24, and iGMS all have different API capabilities and rate push latency

The Core Gap in the Market

Existing RMS tools (PriceLabs, Beyond Pricing, Wheelhouse, DPGO) solve the algorithm problem — not the execution problem. They produce recommendations. Someone still has to review, decide, push, verify, and watch. For a property manager running 15–50 units, this is a 2–4 hour daily task that gets skipped, batched, or delegated to inexperienced staff. This system eliminates that execution burden while keeping humans in control of strategy.

1.2 What This System Is — and Is Not

This system IS	This system IS NOT
A daily execution engine that translates revenue signals into verified PMS actions	A replacement for PriceLabs, Beyond Pricing, or Wheelhouse (it can consume their outputs as signals)
A supervised AI that proposes, explains, and waits for approval on non-trivial changes	A strategy consultant or long-term revenue planner (V1 scope)
A monitoring layer that detects post-execution drift and rolls back automatically	A fully autonomous pricing engine — human approval gates are non-negotiable in V1
A reservation triage layer that handles routine guest queries without human involvement	An owner-facing analytics product or listing optimization tool

1.3 Competitive Landscape & Differentiation

Tool	Category	The Gap This System Fills
PriceLabs	Algorithm + dashboard	No execution; no approval workflow; no rollback
Beyond Pricing	Algorithm + autopilot	Autopilot is opaque; no explainability; no rollback

Tool	Category	The Gap This System Fills
Wheelhouse	Algorithm + strategy	Strategy-first; no operational execution layer
DPGO	Algorithm (Dubai-aware)	Limited PMS integrations; no agentic layer
Hostaway/Guesty native	Basic rules engine	Not signal-aware; no ML; no gap/LOS intelligence

Our Core Differentiation

Execution fidelity + explainability + human control + verified rollback. We are the operating system for revenue execution — not another pricing algorithm.

PART 2: TARGET USERS & USE CASES

2.1 Primary User Personas

Property Manager (15–50 units, self-managing or small team)

- Pain: Spends 2–3 hours/day on pricing, inbox, and calendar management
- Goal: Cut that to <30 minutes review time while maintaining yield
- Usage pattern: Morning review of overnight proposals; approves batch; monitors during day

Revenue Manager (in-house, managing a single portfolio or multiple)

- Pain: Context-switching between PMS, pricing tool, calendar, and competitor data
- Goal: Consolidated view, faster execution cycle, fewer missed opportunities
- Usage pattern: Deep morning session; uses CRO chat to interrogate decisions; spot-checks anomalies

Revenue Management Agency (managing 5–15 client portfolios)

- Pain: Scaling is headcount-constrained; junior staff make inconsistent decisions
- Goal: Systematize execution so senior RMs focus on strategy, not daily operations
- Usage pattern: Portfolio-level oversight; per-client approval workflows; audit trails for clients

2.2 User Roles & Permissions

Role	Permissions	Approval Authority
Admin	Full configuration, state control, user management	All actions including high-risk
Revenue Manager	Review proposals, approve/reject, view all data	Medium and high risk
Operator (Ops)	View activity, manage inbox escalations	None — view only
Read-Only	Audit trail, summary dashboards only	None

2.3 Core Use Cases (V1 Scope)

#	Use Case	Description
1	Daily pricing execution	System reviews portfolio daily, proposes changes, executes on approval
2	Gap night fill	Detects orphan nights and proposes LOS adjustments or targeted discounts to capture bookings
3	Event-driven surge pricing	Detects confirmed events, proposes conservative price lifts within guardrails
4	Competitor compression response	When competitive set goes unavailable, flags demand signal and proposes lift

#	Use Case	Description
5	Post-booking anomaly detection	Detects unexpected calendar changes post-execution and pauses for review
6	Reservation inbox triage	Handles routine guest inquiries; escalates sensitive cases to human operators
7	Rollback on failure	Any execution that doesn't verify correctly is automatically and verifiably reversed

PART 3: SYSTEM STATES

3.1 State Definitions

The system always exists in exactly one of five states. State governs what is permitted. No execution can bypass state enforcement.

State	Description	Execution	Proposals	Monitoring
Connected	PMS linked, credentials verified, no operations started	✗	✗	✗
Observing	Read-only; system ingests data, builds baseline	✗	✗	✓
Simulating	Proposals visible; user can inspect, test guardrails	✗	✓	✓
Active	Full execution enabled under approval workflow	✓	✓	✓
Paused	Hard stop; no execution; anomalies or user-triggered	✗	✗	✓

3.2 State Transition Rules

Permitted Transitions

- Connected → Observing (after baseline data pull completes)
- Observing → Simulating (manual progression by Admin)
- Simulating → Active (explicit human activation — always requires human intent)
- Active → Paused (anomaly detection or user command)
- Observing → Paused (data integrity failure)
- Simulating → Paused (guardrail breach detected)
- Paused → Observing (manual resume by Admin only)
- Active → Simulating (manual demotion by Admin)

Forbidden Transitions (hardcoded — cannot be overridden)

- Connected → Active (must progress through all stages)
- Connected → Simulating (no stage skipping)
- Paused → Active (cannot resume directly to execution mode)
- Any automated path → Active (human intent required every time)

3.3 Pause Taxonomy

Not all pauses are equal. The system classifies pause triggers to determine the correct resume path.

Pause Type	Trigger	Resume Path
Anomaly Pause	Unexpected post-execution state detected	Admin reviews anomaly, confirms safe, manual resume
Integrity Pause	Stale or inconsistent PMS data	Data Aggregator confirms fresh data; Admin resumes
Guardrail Pause	Proposal exceeds all configured bounds	Config review required; Admin resumes
User Pause	Explicit human command via CRO chat or button	Admin resumes at any time
Rate Limit Pause	PMS API throttle hits limit	Auto-wait 15 min; resume if cleared; escalate if >1 hr

PART 4: AGENT ARCHITECTURE

4.1 Architectural Principles

Core Design Rules

1. One Manager Agent (CRO) — owns orchestration, state, user interaction, and execution authority
2. Multiple Worker Agents — each performs a narrow, bounded task; never executes autonomously
3. Workers produce signals or proposals; only CRO can authorize execution
4. Only Channel Sync Agent writes to PMS; all other agents are read-only
5. Workers never call each other directly; all routing goes through CRO
6. Every agent interaction is logged with inputs, outputs, rationale, and timestamp

4.2 Manager Agent: Chief Revenue Officer (CRO)

The CRO is the single authority and orchestration layer. It is a conversational AI agent — the only point of contact for users and the only point of authority for all agent orchestration and execution decisions.

Core Responsibilities

- Maintain and enforce system state across all agents
- Schedule and trigger worker agents based on configured cycles
- Consolidate worker outputs into structured, human-readable proposals
- Determine risk classification for each proposal batch
- Present proposals with plain-language explanations ("Why is this being recommended?")
- Collect approval from users when required; authorize execution when approved
- Monitor Anomaly Detector alerts and trigger pause when required
- Maintain complete, queryable audit trail for all decisions and outcomes

CRO Interface Modes

Mode	Trigger	Description
Proposal Mode	Morning briefing	Queued proposals sorted by urgency and risk level; approval cards
Interrogation Mode	User question	"Why is unit 12 priced at AED 680?" — CRO explains decision chain
Audit Mode	User request	Reviews past N days of actions, outcomes, and rollbacks with full detail
Alert Mode	Anomaly/failure	Urgent notifications about pauses, execution failures, anomalies

CRO Constraints (V1 — Non-Negotiable)

- Cannot bypass Adjustment Reviewer veto under any circumstance
- Cannot execute in any state other than Active
- Cannot write directly to PMS — all writes go through Channel Sync Agent

- Cannot modify its own guardrail configuration at runtime
- Cannot self-trigger execution — all execution requires approval or auto-approve criteria

4.3 Worker Agents

Agent 1: Data Aggregator

Role: Single source of truth for all portfolio data. Nothing enters the system without passing through this agent. Downstream agents receive only normalized, freshness-flagged snapshots.

Inputs: PMS credentials and property mappings; scheduled + on-demand refresh triggers from CRO

Outputs: Normalized portfolio snapshot (calendars, prices, bookings, LOS rules, blocked dates, channel mappings); freshness timestamps; data consistency flags; change deltas

Key Intelligence

- Detects OTA calendar desync: Airbnb calendar vs PMS calendar divergence
- Flags channel manager push latency exceeding 30 minutes
- Identifies price inconsistency across OTA channels for same unit/date
- Marks dates as "unconfirmed" when booking data is incomplete

> Constraint: Read-only access to PMS. Must flag stale data (>8 hours old) and require fresh pull before any execution cycle. Never passes stale data to downstream agents without explicit staleness flag.

Agent 2: Event Intelligence Agent

Role: Monitors external event signals that drive demand in Dubai. Conservative by design — prioritizes avoiding false positives that lead to overpricing and missed bookings over capturing every demand spike.

Inputs: Configured event feeds (Dubai Calendar, DTCM, Time Out Dubai, Eventbrite, venue websites); geographic filters per property zone; confidence threshold configuration

Outputs: Event signals with: event name, date range, zone, confidence score (Low/Medium/High), demand radius, historical comparison, and plain-language rationale

Dubai Event Calendar the Agent Must Track

Event	Timing	Zone Impact	Demand Type
UAE National Day	Dec 2–3	City-wide compression	Staycation + GCC visitors
New Year's Eve	Dec 31	Downtown, Palm, Marina premium	Leisure; extremely high ADR potential
Dubai Shopping Festival	Jan–Feb	Zone-specific; retail corridors	Family leisure; GCC drive market
GITEX Global	Oct (5 days)	DIFC, Business Bay, Media City	Corporate; high ADR; long stays
Art Dubai / Dubai Art Week	March	DIFC, Downtown	Premium leisure; international
Dubai World Cup (horse racing)	March	City-wide leisure spike	International; premium segment

Event	Timing	Zone Impact	Demand Type
Dubai International Film Festival	December	DIFC, Downtown	International; premium leisure
Ramadan Onset / Eid Al Fitr	Variable	Character shift; not demand drop	Different segments; different LOS
Eid Al Adha	Variable (June/July)	Often underestimated	GCC family drive market spike
F1 Abu Dhabi (spillover)	November	Marina, Downtown, JBR	International premium overflow
Dubai Fitness Challenge	Oct–Nov (30 days)	Active lifestyle zones	Health/wellness segment

Constraint: No pricing recommendations — outputs signals only. Never outputs "High" confidence for unverified events. Conservative confidence is the default — would rather miss a 5% lift than cause a vacancy-inducing overpricing spike.

Agent 3: Competitor Scanner

Role: Monitors the competitive set for each property to detect availability compression (demand signal) and price movements (market signal). Think of this as real-time market intelligence without direct OTA scraping.

Inputs: Competitive set definitions per property (5–10 comparable units by zone, bedroom count, amenity tier); scan schedule; target OTA channels (Airbnb, Booking.com primary in Dubai)

Outputs: Compression signals (% of comp set unavailable for date range); price positioning signals (our price vs market median); compression trend direction; market tier analysis

Signal Intelligence

- Distinguishes temporary blocks (owner blocks, maintenance holds) from genuine bookings using multi-scan trend analysis
- Filters noise: a single property unavailable is not a signal; 60%+ of competitive set unavailable is a High signal
- Detects price floor movements: if competitive set is raising minimums, market is tightening
- Identifies LOS compression: if competitors are all rejecting 1-night stays, minimum stay demand is shifting

Constraint: No pricing decisions — outputs market signals only. Requires trend confirmation across 2+ scans for High confidence to prevent false positives from temporary competitor blocks.

Agent 4: Pricing Optimizer

Role: The analytical core. Consumes all signals and portfolio data to propose specific price changes for specific units on specific dates. All proposals are classified, risk-rated, and explained before being passed to the Adjustment Reviewer.

Inputs: Portfolio snapshot, event signals, competitor signals, PM strategy parameters (target occupancy, base price floors/ceilings, seasonal curves, risk appetite), historical booking data

Date Classification System

Every date in the booking horizon receives exactly one classification:

Classification	Definition	Typical Action
Protected	Confirmed high-value events, historically strong dates, or >70% portfolio already occupied	Hold or increase; never discount under any circumstance
Healthy	Normal demand, >45 days out, current price appropriate	Monitor; minor adjustments only; no urgency
At Risk	<30 days out, below target occupancy, no strong demand signal	Targeted price reduction within guardrails
Distressed	<14 days out, vacant, no bookings, no strong demand signal	Aggressive but bounded discounting; LOS relaxation

Dubai Market Intelligence Encoded

- Weekend premium dynamics: Thursday/Friday nights (Islamic weekend) command premiums vs Mon–Wed
- Ramadan pricing: volume shifts, different lead times, cultural demand character changes
- Summer demand trough (July–August): accelerated discounting required; cooling-off compression effect
- School holiday premium: UK, India, Russia school calendars drive distinct family segment spikes
- Corporate/transit demand: DIFC/Business Bay units see different midweek corporate patterns vs Marina leisure units

Constraint: Proposals only; never executes. Must assign risk level to every proposal. Must not generate conflicting proposals for the same unit/date. Must obey configured floor and ceiling for every unit.

Agent 5: Gap & LOS Optimizer

Role: Solves the orphan night problem — the single most common and consistently underaddressed revenue leak in short-term rental portfolios. A single-night gap between two bookings is effectively dead inventory without proactive LOS management.

Gap Detection & Resolution Logic

Gap Type	Condition	Primary Fix	Secondary Fix
1-night orphan	Single night between two bookings	Reduce min stay to 1 night	Apply gap fill discount if needed
2-night micro-gap	2 nights available with 3-night min	Reduce min stay to 2 nights	Light discount if conversion low
3-night gap	3 nights with 5-night min	Reduce min stay to 3 nights	Small discount if 48hrs to arrival
Last-minute vacancy	<5 days to check-in, empty nights	LOS relaxation + last-minute discount	Same-night pricing if applicable
LOS-demand mismatch	Unit has 7-night min, demand is 2–3 nights	Temporary LOS reduction for date range	Return to standard LOS after period

Auto-Revert Intelligence

- If gap goes unfilled 48 hours before the orphan date, revert to standard LOS (avoids attracting misaligned guest profiles)
- If discount is applied and unit books, cancel the pending revert trigger immediately
- All revert events logged for portfolio-level LOS optimization learning

> LOS changes always take priority over discount changes — solve the gap with LOS relaxation first, apply discount only if LOS change alone is insufficient. Never modify LOS on Protected dates. Never create a gap discount that conflicts with an adjacent booking constraint.

Agent 6: Adjustment Reviewer

Role: The guardrails enforcer. Independent validation layer between proposal generation and CRO presentation. Cannot be influenced by other agents. The CRO cannot override its veto in V1.

Approval Classification Rules

Criteria	Risk Level	Approval Required?
Price change <5%, no event context, within guardrails	Low	No — auto-approve
Price change 5–15%, single signal, within guardrails	Medium	Yes — human approval
Price change >15%, any level	High	Yes — human approval
Event-driven adjustment, any size	Medium/High	Yes — human approval

Criteria	Risk Level	Approval Required?
Conflicting signals detected	High	Yes — human approval
Multiple rule conflicts in same batch	High	Yes — human approval
Any Protected date modification	High	Yes — human approval
Gap fix within configured auto-approve threshold	Low	No — auto-approve

Veto Authority: The Adjustment Reviewer has unconditional veto power in V1. The CRO cannot override a veto. Vetoed proposals are presented to the user with the specific veto reason if the user asks for explanation. Cannot be instructed to soften guardrails mid-session.

Agent 7: Channel Sync Agent

Role: The execution layer. The only agent that writes to PMS. Every write is verified against a post-execution snapshot. Every verified write stores a rollback payload that remains available for 72 hours.

Execution Preconditions (All Must Pass Before Any Write)

- ☐ System state = Active
- ☐ Data Aggregator snapshot freshness < 30 minutes
- ☐ Human approval obtained (Medium/High) OR auto-approve criteria confirmed (Low)
- ☐ Adjustment Reviewer approval on file — not vetoed
- ☐ Rollback payload constructed and stored in persistent log
- ☐ PMS API health check: last successful API call within 5 minutes
- ☐ No active Critical anomalies in portfolio

Execution & Verification Process

1. Construct API payload for target PMS format
2. Store pre-execution state as rollback payload (persisted)
3. Execute API call to PMS
4. Wait for PMS confirmation (timeout: 30 seconds)
5. Pull post-execution PMS snapshot for changed unit/date range
6. Compare expected state vs actual state within tolerance thresholds
7. If match: log success, mark rollback payload as "available"
8. If mismatch: trigger immediate rollback, log failure, notify CRO for escalation

Verification Standards

- Price changes: post-execution price must match proposed price within rounding tolerance (±1 AED)
- LOS changes: minimum stay must reflect new LOS rule within 5 minutes of execution
- Gap discounts: promotional price must be active on all target dates
- OTA propagation: calendar and price sync must propagate to Airbnb/Booking.com within 15 minutes (flag if not, escalate if >30 minutes)

Agent 8: Anomaly Detector

Role: Post-execution watchdog. Monitors outcomes against expectations continuously. The only agent with authority to recommend system pause — requests go through CRO, not directly to state machine.

Anomaly Taxonomy

Anomaly	Description	Severity	Response
Price drift	Price changed unexpectedly after execution (channel manager override, OTA correction)	Warning	Alert CRO; propose rollback for review
Calendar desync	Airbnb calendar and PMS show different availability	Critical	Alert CRO; pause execution immediately
Unexpected booking	Booking received at non-standard price (possible OTA pricing override)	Warning	Alert CRO; flag for investigation
Rollback failure	A rollback did not successfully restore previous state	Critical	Immediate escalate to Admin; pause system
PMS API degradation	Execution success rate dropping below 95% over rolling hour	Warning	Reduce batch size; escalate if >30 min
Double booking risk	Calendar conflicts detected post-execution	Critical	Immediate pause; escalate to Admin

Monitoring Schedule

- T+5 minutes after execution: initial verification
- T+30 minutes: OTA propagation check
- T+120 minutes: full post-execution health check
- Continuous: booking event stream from PMS webhooks

Agent 9: Reservation Agent

Role: Handles the PMS unified inbox for routine guest communication. Completely isolated from the revenue decision loop in V1. Operates in parallel with the revenue cycle and does not influence pricing under any circumstance.

Handles Automatically

- Check-in and check-out time questions
- WiFi password and access code requests (post-confirmation only)
- Parking instructions and building access details
- Amenity questions: pool, gym, laundry, kitchen equipment
- Early check-in / late check-out availability checks (checks calendar; does NOT approve or quote price)
- Booking confirmation acknowledgments

Always Escalates to Human Operator

- Complaints or disputes of any kind

- Damage reports
- Refund requests
- Extension requests requiring pricing decisions
- Any message with negative sentiment or urgency language (detected via NLP)
- Booking modification requests of any kind
- Any message from a guest with a prior dispute flag in their history

V1 Constraint: No pricing authority. No interaction with revenue agents. Cannot send messages without confirming sender is a verified, confirmed guest. Does not handle pre-booking inquiries in V1.

PART 5: ORCHESTRATION FLOW

5.1 Core Revenue Loop

Triggered daily at configurable time (default: 6:00 AM Dubai time). Can be triggered on-demand via CRO chat for specific units or date ranges.

Step	Agent	Action
1	Data Aggregator	Pulls fresh PMS snapshot; flags data issues; confirms freshness before proceeding
2	Event Intelligence Agent	Refreshes event signals for next 90-day horizon; updates confidence scores
3	Competitor Scanner	Scans competitive set for availability compression and price movements
4	Pricing Optimizer	Consumes all signals; classifies all dates; generates price change proposals with risk ratings
5	Gap & LOS Optimizer	Reviews calendar for orphan nights and LOS mismatches; generates optimization proposals
6	Adjustment Reviewer	Validates all proposals against guardrails; classifies risk; vetoes unsafe proposals
7	CRO	Consolidates approved proposals into morning briefing; queues approval cards for user
8	User Review	Reviews Medium/High risk proposals; approves, rejects, or modifies batch; Low risk auto-executes
9	CRO (Authorization)	Authorizes Channel Sync for all approved actions; logs authorization event
10	Channel Sync Agent	Executes with pre-execution state capture; verifies success; stores rollback payloads
11	Anomaly Detector	Monitors post-execution at T+5, T+30, and T+120 minutes; alerts on unexpected states
12	CRO (Summary)	Delivers daily summary at configurable time (default 8:00 PM Dubai time)

5.2 Parallel Reservation Loop

The Reservation Agent operates continuously and independently of the revenue cycle. It monitors the PMS unified inbox, responds to routine queries automatically, and escalates complex cases to human operators. It has zero interaction with revenue agents in V1.

5.3 Ad-Hoc Loops (User-Triggered via CRO Chat)

- "Re-run pricing for next weekend" — triggers on-demand revenue loop for specific date range on specific units
- "What does our competitive set look like for NYE?" — triggers Competitor Scanner + Event Intelligence on demand

- "Why did price on Unit 7 drop?" — CRO interrogates decision logs and explains complete decision chain
- "Pause everything while I check the Airbnb calendar" — triggers User Pause state immediately
- "Undo the pricing changes from this morning" — triggers user-initiated rollback for specified execution batch

PART 6: CONFIGURATION & GUARDRAILS

6.1 Per-Portfolio Configuration

Parameter	Description	Default
Execution schedule	Time daily revenue loop runs	6:00 AM Dubai (GST)
Pricing horizon	How many days ahead to optimize	90 days
Base price (per unit)	Absolute floor; never propose below this	PM-defined at setup
Price ceiling (per unit)	Absolute maximum; never propose above this	PM-defined at setup
Max single-day change	Maximum % change per pricing cycle	15%
Max weekly drift	Maximum % from base over rolling 7 days	40%
Auto-approve threshold	Low-risk changes below this % execute without human review	5%
Event confidence minimum	Minimum confidence level to trigger event-based pricing	Medium
Gap fill discount max	Maximum discount for orphan night fills	20%
Competitor compression threshold	% of comp set unavailable before flagging signal	60%
Rollback retention window	How long rollback payloads are retained	72 hours
PMS sync latency tolerance	Max acceptable delay for OTA calendar propagation	15 minutes

6.2 Strategy Modes (V1 Simplified Onboarding)

V1 exposes three pre-configured strategy modes to reduce cognitive load during onboarding. Strategy mode sets all portfolio-level parameters; PM can still override individual units.

Mode	Philosophy	Risk Appetite	Best For
Conservative	Maximize occupancy; minimize vacancy risk	Low max change; high auto-approve threshold	New operators; high-competition zones; lower ADR markets
Balanced	Balance occupancy and ADR; standard execution cadence	Moderate max change; standard auto-approve	Most PM portfolios; established operators
Aggressive	Maximize ADR; tolerate some vacancy for yield uplift	High max change; low auto-approve threshold	Premium properties; low-competition zones; experienced RMs

PART 7: EXECUTION & ROLLBACK

7.1 Execution Precondition Checklist

Before any write to PMS, all of the following must pass. If any check fails, execution halts and CRO notifies user. System does not auto-pause unless the failure is an active anomaly.

- ☐ System state = Active (enforced at state machine level)
- ☐ Data freshness < 30 minutes (confirmed by Data Aggregator timestamp)
- ☐ Approval obtained: human approval for Medium/High, or auto-approve criteria met for Low
- ☐ Adjustment Reviewer approval on record — proposal not in vetoed state
- ☐ Rollback payload constructed, stored, and confirmed retrievable
- ☐ PMS API health: last successful API call within past 5 minutes
- ☐ No active Critical anomalies in the affected portfolio

7.2 Rollback Design

Rollback Payload Contents

- Unit identifier and all channel mappings
- Pre-execution prices for each date in the changed range
- Pre-execution LOS rules for each affected date
- Pre-execution promotional/discount state
- Capture timestamp and PMS API endpoint + payload format for reversal
- Verification fingerprint: what the state looked like before execution

Rollback Triggers

Trigger Type	Origin	Response Time
Channel Sync verification failure	Automatic — immediate	< 30 seconds from failure detection
Anomaly Detector Critical alert	After CRO review and authorization	Within 5 minutes of alert
Calendar desync detected post-execution	Automatic — immediate	< 30 seconds from detection
"Undo last execution" command	User-initiated via CRO chat	Within 2 minutes of command
Specific unit/date rollback	User-initiated via CRO chat	Within 2 minutes of command

Rollback Guarantees

- Every executed action has a rollback payload — no exceptions
- Rollback payloads retained for 72 hours after execution
- Rollback success must be verified using the same standard as original execution
- Failed rollbacks escalate immediately to Admin via all channels (WhatsApp, Slack, platform) — no exceptions

PART 8: NOTIFICATIONS & COMMUNICATION

8.1 Primary Channel: CRO Chat

All routine interaction happens through the CRO conversational interface. This is not a notification feed — it is an interactive channel where users can ask questions, challenge proposals, give commands, and interrogate past decisions.

Cadence	Default Time	Content
Morning briefing	7:00 AM Dubai	Proposal queue summary, market intelligence highlights, overnight event signals
Approval cards	After briefing	Structured approval cards for each proposal batch requiring human review
On-demand interrogation	Any time	Answer questions about any proposal, unit, date, or past decision in natural language
Evening summary	8:00 PM Dubai	Executions completed, bookings received since morning, anomalies resolved or active
Audit thread	Always accessible	Full history of every decision, execution, and outcome in queryable format

8.2 Secondary Channels: WhatsApp & Slack

Used only for urgent, time-sensitive alerts where a delay of more than 5 minutes is unacceptable. These channels are not for routine communication.

Event	Channels	Timing
Critical anomaly detected	WhatsApp + Slack	Immediate
Execution failure requiring manual intervention	WhatsApp + Slack	Immediate
System paused (anomaly-driven)	WhatsApp + Slack	Immediate
Rollback failure	WhatsApp + Slack + CRO	Immediate
High-risk proposal awaiting approval (configurable SLA)	Slack	After SLA threshold (default: 30 min)

> Explicitly NOT sent to WhatsApp/Slack: routine proposal notifications, low-risk auto-approved executions, daily summaries, informational alerts. Secondary channels are reserved for emergencies only.

PART 9: OBSERVABILITY & AUDIT

9.1 Required Log Fields (Every Action)

Every decision, proposal, and execution generates a log entry. No action can be unlogged.

Field	Description
Action ID	UUID — unique identifier for every log entry
Timestamp	UTC and Dubai local time (GST)
Agent	Which agent generated this action
System State	System state at the time of the action
Input Summary	Which signals and data contributed to this decision
Decision Rationale	Plain-language explanation of why this action was proposed or taken
Risk Classification	Low / Medium / High with specific criteria that determined the level
Approval Record	Auto-approved (with criteria) OR human-approved (user ID, timestamp) OR vetoed (reason)
Execution Result	Success / Failure / Skipped (with reason for skip or failure)
Rollback Status	Available / Triggered / Completed / Expired
Verification Outcome	Matched / Mismatched / Timeout (with details on mismatch if applicable)

9.2 Audit Query Interface

Users query the audit log through natural language in the CRO chat interface:

- "Show me everything that happened to Unit 12 last week"
- "Why did we rollback the Marina Palm execution on Thursday?"
- "What was our ADR change versus baseline since we went Active?"
- "How many proposals were auto-approved vs human-approved this month?"
- "Which units had anomalies in the last 30 days?"

Audit data must be exportable to CSV for client reporting (Revenue Management Agency use case).

PART 10: PMS INTEGRATION ARCHITECTURE

10.1 V1 Target PMS Integrations

Priority order based on Dubai SMB property manager market share:

Priority	PMS	Dubai Market Share	API Maturity	Notes
1	Hostaway	Largest SMB share	Excellent	Full API for pricing, calendar, inbox; webhook support
2	Guesty	Mid-market operators	Excellent	Comprehensive API; used by larger operators
3	Lodgify	Growing share	Good	API maturity acceptable for V1 requirements
4	Beds24	Legacy install base	Functional	Older API; significant installed user base in Dubai
5	Hostfully	Growing	Good	Growing Dubai adoption; clean API

10.2 Required API Capabilities

Capability	Required?	Notes
Read calendar and booking data	Required	Core; execution depends on fresh calendar data
Read current prices per channel	Required	Core; cannot operate without this
Write prices to PMS	Required	Core execution capability
Write LOS rules	Required	Required for gap and LOS optimization
Booking event webhooks	Required	Required for real-time anomaly detection
Unified inbox read access	Required	Required for Reservation Agent
Unified inbox write (responses)	Required	Required for Reservation Agent auto-response
Price change event webhooks	Preferred	Enables faster post-execution verification
Rate plan / promotion management	Preferred	Enables more sophisticated gap fill tactics in V2

10.3 Channel Manager Considerations

Many Dubai property managers use a channel manager layer (Rentals United, Cloudbeds, Hostaway built-in) between PMS and OTAs. The system must:

- Account for channel manager propagation latency to Airbnb and Booking.com (typically 5–20 minutes)
- Verify at the OTA level (not just PMS level) for high-risk executions
- Flag when PMS-to-OTA sync is broken or delayed beyond tolerance thresholds
- Never count a High-risk execution as verified until OTA-level confirmation is received

PART 11: SUCCESS CRITERIA

11.1 Functional Success Criteria

Criterion	Measurement	Target
Price execution accuracy	Post-execution verification match rate	>95%
Rollback reliability	Successful rollback rate when triggered	100%
Pause response time	Time from trigger to full halt	<30 seconds
Reservation auto-handle rate	% of inbox messages resolved without human	>80%
Daily cycle automation	Consecutive days without manual intervention	5+ days
Zero out-of-state execution	Executions occurring outside Active state	0 (absolute)

11.2 Behavioral Success Criteria

Criterion	Measurement	Target
Daily management time saved	PM time on pricing review vs baseline	<30 min vs 2–3 hr prior
Proposal acceptance rate	% of proposals approved without modification	>85%
Rollback frequency	% of total executions that trigger rollback	<5%
Gap fill improvement	Orphan night fill rate vs 90-day baseline	Measurable improvement in 30 days
ADR impact	Average daily rate vs manual baseline	Neutral or positive in 60 days

11.3 Trust Criteria (Qualitative)

The system has succeeded in V1 when:

- Users feel informed about every action the system takes — no surprises
- Users feel in control — they can pause, undo, or override at any time with one command
- Users trust the system to be correct when verified — they stop double-checking every change
- When the system makes a mistake, it is visible and recoverable — trust is preserved even under failure

PART 12: V2 / V3 FORWARD REFERENCES

V2 Planned Expansions

Feature	Description
Strategy Layer	Long-term revenue planning; seasonal strategy design; target ADR/occupancy modeling per unit type
Portfolio-level optimization	Cross-unit pricing considering cannibalization and complementarity; portfolio yield management
Owner-facing analytics	Separate read-only dashboard for property owners with performance context
Enhanced learning	System learns from booking outcomes to improve proposal accuracy and reduce veto rate
Multi-market support	Expansion from Dubai to Abu Dhabi, Ras Al Khaimah, Riyadh, and international markets
RMS signal integration	Consume PriceLabs or Beyond Pricing recommendations as one signal source alongside native signals
Demand forecasting	ML-based demand curves per unit type per season; booking window prediction

V3 Planned Expansions

Feature	Description
Partial autonomy mode	Trusted execution within very narrow parameters without human approval; highly configurable; opt-in only
Listing optimization agent	Suggests photo, title, description, and amenity highlight changes based on search performance data
Revenue strategy agent	Proactive recommendations for portfolio composition changes; unit-level repositioning
Dynamic LOS pricing matrix	Price-per-length-of-stay optimization; discount structure per LOS bracket
Predictive guest quality scoring	Risk-assess booking enquiries before acceptance based on profile signals

PART 13: DEFINITION OF DONE (V1)

V1 is complete when all of the following are true. These are binary criteria — no partial credit.

#	Criterion	How Verified
1	System completes a full daily revenue cycle without human babysitting	Automated test: 5+ consecutive days of uninterrupted operation
2	Every execution is preceded by a logged, approved proposal	Audit query: 0 executions without prior approval log entry
3	No execution ever occurs outside Active state	State machine log: 0 execution events in non-Active states
4	Rollback works reliably in all tested failure scenarios	Test suite: inject failures; verify 100% rollback success rate
5	Reservation Agent meaningfully reduces inbox workload	Measure: >80% routine query auto-resolution in live pilot
6	At least one live portfolio runs Active for 14 consecutive days	Live pilot: no manual interventions required for standard operations
7	CRO can explain any past decision in plain language	User test: CRO successfully answers 10 random audit questions
8	Audit trail is complete, queryable, and exportable	QA: query any past action; export to CSV; no missing entries
9	Admin can pause, resume, and review without engineering support	UAT: Admin completes all operations via UI with no dev help
10	Trust is preserved under failure — mistake is visible and recoverable	Failure simulation: users rate system trustworthiness during recovery